Security analysis of GroupTEE

1 Introduction

In this section, we present a formal security analysis of the proposed protocol in the Random Oracle Model (ROM). Authentication is proven under the Existential Unforgeability against Chosen-message Attacks (EUF-CMA) of the Schnorr signature scheme, while session-key confidentiality is established under standard Diffie-Hellman assumptions in the ROM via a real-or-random (ROR) indistinguishability experiment. The formal proofs are further complemented by symbolic analysis using BAN logic and automated verification with the Scyther tool. In addition, we provide an informal discussion covering replay resistance, forward and backward secrecy, and key agreement consistency.

2 Theoretical Proof

2.1 Authentication Proof

Assumptions. We work in a cyclic group $\mathbb{G}$ of prime order q generated by G. Each participant P_i holds a long-term secret key $sk_i \in \mathbb{Z}_q^*$ and the corresponding public key $pk_i = sk_i \cdot G$. The hash function $\mathsf{H} : \{0,1\}^* \to \mathbb{Z}_q$ is modeled as a random oracle. For the reduction, we assume access to a standard *single-signature Schnorr* signing oracle for a designated target identity, as required by the EUF-CMA security definition.

Honest Non-Colluding Model. We consider an honest-but-non-colluding execution model. All participating entities are assumed to follow the protocol specification faithfully and to generate authentication messages strictly according to the Schnorr signing procedure using fresh randomness and their own secret keys. No participant deviates from the prescribed message-generation algorithm, and no subset of participants colludes with the adversary to manipulate or forge authentication transcripts. Long-term secret keys remain confidential and are never revealed to other participants or to the adversary.

Theorem 1 (EUF-CMA authenticity of first-round messages).
Assume the hash function H *is modeled as a random oracle. If there exists a PPT adversary* $\mathcal{A}$ *that makes some honest participant accept a forged first-round message with non-negligible probability, then there exists a PPT reduction* $\mathcal{F}$ *that breaks the EUF-CMA security of the* single-signature *Schnorr scheme with non-negligible advantage.*

Proof. Define the signed message for m_i as $\mu_i := (ID_i \parallel R_{ix} \parallel R_{iy} \parallel \mathsf{Encode}(ID_{list}) \parallel sid)$, and the corresponding Schnorr signature as $\sigma_i^{(1)} = (R_i, s_i)$.

Game G0 (Real execution). We consider the real protocol execution, where $\mathcal{A}$ may intercept, modify, or inject first-round messages $m_j = (ID_j, R_j, s_j)$.

Game G1 (Embedding a Schnorr EUF-CMA challenge). $\mathcal{F}$ guesses a target identity index $j^\star \leftarrow \{1, \ldots, n\}$. For all $j \neq j^\star$, $\mathcal{F}$ generates (sk_j, pk_j) honestly and simulates the protocol. For the target $j^\star$, $\mathcal{F}$ sets $pk_{j^\star}$ to be the public key from the Schnorr EUF-CMA challenger and answers every protocol-generated signature of the form $\mathsf{Sign}(sk_{j^\star}, \mu)$ using the signing oracle. Hash queries are answered via a random-oracle table.

Game G2 (Extracting a forgery). Suppose $\mathcal{A}$ succeeds in making some honest neighbor accept a first-round message $\hat{m}_{j^\star} = \{ID_{j^\star}, \hat{R}_{j^\star x}, \hat{R}_{j^\star y}, \hat{s}_{j^\star}\}$ that verifies under $pk_{j^\star}$ for the message $\hat{\mu}_{j^\star} = (ID_{j^\star} \parallel \hat{R}_{j^\star x} \parallel \hat{R}_{j^\star y} \parallel \mathsf{Encode}(ID_{list}) \parallel sid)$.

If $\hat{\mu}_{j^\star}$ was never queried to the signing oracle in Game 1, then $(\hat{\mu}_{j^\star}, (\hat{R}_{j^\star}, \hat{s}_{j^\star}))$ is a valid Schnorr EUF-CMA forgery and $\mathcal{F}$ outputs it.

Otherwise, $\hat{m}_{j^\star}$ is merely a replay of a previously issued protocol signature for the same (ID_{list}, sid) and does not constitute a new-message forgery. In that case, $\mathcal{F}$ aborts.

By the standard index-guessing argument,

$$\mathsf{Adv}_{\mathsf{1st\text{-}auth}}(\mathcal{A}) \;\leq\; n \cdot \mathsf{Adv}^{\mathsf{EUF\text{-}CMA}}_{\mathsf{Schnorr}}(\mathcal{F}) \;+\; \mathsf{negl}(\lambda).$$

Since Schnorr is EUF-CMA secure in the ROM [2], $\mathsf{Adv}_{\mathsf{1st\text{-}auth}}(\mathcal{A})$ is negligible.

Theorem 2 (EUF-CMA authenticity of second-round messages under batch verification).

Assume H *is modeled as a random oracle. If a PPT adversary* $\mathcal{A}$ *makes an honest participant* P_i *accept the second-round batch verification in Step 8 with non-negligible probability, then there exists a PPT reduction* $\mathcal{F}$ *that breaks the EUF-CMA security of the* single-signature *Schnorr scheme with non-negligible advantage, except with probability at most* $1/q$ *stemming from randomized batch verification.*

Proof. Define the signed message for M_j as $\nu_j := (ID_j \parallel X_{jx} \parallel X_{jy} \parallel R_{jx} \parallel R_{jy} \parallel \mathsf{Encode}(ID_{list}) \parallel sid)$, and the corresponding Schnorr signature as $\sigma_j^{(2)} = (R_j, t_j)$.

Game G0 (Embedding a Schnorr EUF-CMA challenge). $\mathcal{A}$ interacts with honest parties and controls the network, aiming to make some honest P_i accept Step 8(4).

Game G1 (Oracle setup and ROM programming). $\mathcal{F}$ guesses a target signer index $j^\star \leftarrow \{1, \ldots, n\}$. For all $j \neq j^\star$, $\mathcal{F}$ generates keys honestly and simulates the protocol. For $j^\star$, $\mathcal{F}$ sets $pk_{j^\star}$ to the challenger public key and answers every protocol-generated signature $\mathsf{Sign}(sk_{j^\star}, \nu)$ via the Schnorr signing oracle. Hash queries are answered via a ROM table.

Game G2 (From acceptance to a Schnorr forgery). Suppose $\mathcal{A}$ makes some honest P_i accept Step 8(4), i.e., the randomized batch verification with fresh random weights $\{\alpha_j\}_{j \neq i}$ passes.

Let $\mathcal{E}$ be the event that *there exists* an index $j \neq i$ such that the *individual* Schnorr verification for $(\nu_j, (R_j, t_j))$ fails. By the standard soundness of randomized batch verification [1],

$$\Pr[\mathsf{Accept} \wedge \mathcal{E}] \leq 1/q.$$

Therefore, except with probability at most $1/q$, batch acceptance implies that all individual signatures are valid.

Condition on $\neg\mathcal{E}$. In particular, the individual verification for the target index $j^\star$ is valid, i.e., $(\hat{\nu}_{j^\star}, (\hat{R}_{j^\star}, \hat{t}_{j^\star}))$ verifies under $pk_{j^\star}$. If $\hat{\nu}_{j^\star}$ was never queried to the signing oracle, $\mathcal{F}$ outputs it as an EUF-CMA forgery. Otherwise it is a replay for the same message and $\mathcal{F}$ aborts.

Thus,

$$\mathsf{Adv}_{\mathsf{2nd\text{-}auth}}(\mathcal{A}) \;\leq\; n \cdot \mathsf{Adv}^{\mathsf{EUF\text{-}CMA}}_{\mathsf{Schnorr}}(\mathcal{F}) \;+\; \frac{1}{q} \;+\; \mathsf{negl}(\lambda).$$

Since Schnorr is EUF-CMA secure in the ROM, $\mathsf{Adv}_{\mathsf{2nd\text{-}auth}}(\mathcal{A})$ is negligible.

2.2 Key Consistency Proof

We prove that all honest participants derive the identical group element prior to key derivation.

Theorem 3 (Key consistency).
Under honest execution of the protocol, all honest participants compute the same group element, i.e., for all i, j, $K_i = K_j$.
Consequently, since KDF *is deterministic, $GK_i = GK_j$.*

Proof. Recall that Step 9 defines

$$K_i = nr_iR_{i-1} + (n-1)X_i + (n-2)X_{i+1} + \cdots + X_{i-2},$$

where

$$X_j = r_j(R_{j+1} - R_{j-1}) \quad \text{and} \quad R_j = r_jG.$$

Substituting the definition of X_j into K_i and rearranging terms yields

$$\begin{aligned}
K_i =& nr_i \cdot R_{i-1} + (n-1) \cdot X_i + (n-2) \cdot X_{i+1} + \cdots + X_{i-2} \\
=& r_i \cdot R_{i-1} + (r_i \cdot R_{i-1} + X_i) + (r_i \cdot R_{i-1} + X_i + X_{i+1}) + \cdots + \\
& (r_i \cdot R_{i-1} + X_i + \cdots + X_{i-2}) \\
=& r_i \cdot R_{i-1} + (r_i \cdot R_{i-1} + r_i \cdot R_{i+1} - r_i \cdot R_{i-1}) + \\
& (r_i \cdot R_{i-1} + r_i \cdot R_{i+1} - r_i \cdot R_{i-1} + r_{i+1} \cdot R_{i+2} - r_{i+1} \cdot R_i) + \cdots + \\
& (r_i \cdot R_{i-1} + r_i \cdot R_{i+1} - r_i \cdot R_{i-1} + \cdots + r_{i-2} \cdot R_{i-1} - r_{i-2} \cdot R_{i-3}) \\
=& r_i \cdot R_{i-1} + r_i \cdot R_{i+1} + r_{i+1} \cdot R_{i+2} + \cdots + r_{i-2} \cdot R_{i-1} \\
=& r_i \cdot R_{i+1} + r_{i+1} \cdot R_{i+2} + \cdots + r_{i-2} \cdot R_{i-1} + r_{i-1} \cdot R_i
\end{aligned}$$

Thus,

$$K_i = r_i R_{i+1} + r_{i+1} R_{i+2} + \cdots + r_{i-2} R_{i-1} + r_{i-1} R_i,$$

where indices are taken modulo n.

The right-hand side is a cyclically invariant symmetric sum of adjacent Diffie-Hellman products. Hence it is independent of the index i, and therefore all honest participants compute the same group element.

Since the session key is derived as

$$GK = \mathsf{KDF}(\mathsf{EncodePoint}(K_i)),$$

and KDF is deterministic, all honest parties derive the identical session key.

We prove that all honest participants derive the identical group element K prior to key derivation.

2.3 Confidentiality Proof

Theorem 4 (Session-key indistinguishability under DDH).
Model H *and* KDF *as random oracles. Under the Decisional Diffie-Hellman (DDH) assumption in* $\mathbb{G}$*, the protocol achieves real-or-random (ROR) indistinguishability of the group session key* GK*. Formally, for any PPT adversary* $\mathcal{A}$*,*

$$\mathsf{Adv}_{\mathcal{A}}^{ROR}(\lambda) \leq \mathsf{negl}(\lambda).$$

Proof. We prove the theorem via a sequence of games $G0$–$G3$.

Game $G0$ **(Real execution).** This is the real protocol under the standard ROR experiment. The adversary $\mathcal{A}$ interacts with honest parties and is given either the real session key

$$GK = \mathsf{KDF}(\mathsf{EncodePoint}(K_i))$$

or a uniformly random string of the same length. Let $p_0 = \Pr[b' = 1]$.

Game $G1$ **(ROM simulation).** All hash queries to H and KDF are answered via lazy sampling. Schnorr signatures are simulated consistently in the random oracle model. This changes the adversary's view only negligibly:

$$|p_0 - p_1| \leq \mathsf{negl}(\lambda).$$

Game $G2$ **(DDH embedding).** From Section 2.2, the group element admits the equivalent cyclic representation

$$K_i = r_i R_{i+1} + r_{i+1} R_{i+2} + \cdots + r_{i-1} R_i.$$

Let $(G, A = aG, B = bG, C)$ be a DDH challenge, where $a, b \xleftarrow{\$} \mathbb{Z}_q^*$ and C is either abG or uniformly random in $\mathbb{G}$.

We embed the challenge into the tested session as follows:
(1) Set the ephemeral secret of P_i to $r_i := a$, thus $R_i := A = aG$.
(2) Set $R_{i+1} := B = bG$.
(3) All other ephemeral values are generated honestly.

Under this embedding, the term $r_i R_{i+1}$ equals abG. Define a modified group element

$$\widetilde{K}_i = C + r_{i+1}R_{i+2} + \cdots + r_{i-1}R_i.$$

If $C = abG$, then $\widetilde{K}_i$ is distributed identically to K_i. If C is uniform in $\mathbb{G}$, then $\widetilde{K}_i$ is computationally indistinguishable from uniform, since adding an independent uniform group element masks the remaining independently distributed terms.

Thus any distinguisher between the two cases yields a DDH distinguisher. By the DDH assumption,

$$|p_1 - p_2| \leq \mathsf{negl}(\lambda).$$

Game $G3$ (Random session key). In the random-C branch, $\widetilde{K}_i$ is indistinguishable from uniform. Since KDF is modeled as a random oracle, the derived key is uniformly random unless $\mathcal{A}$ queries the random oracle at the exact input $\mathsf{EncodePoint}(\widetilde{K}_i)$. This event occurs with negligible probability.

Hence,

$$|p_2 - p_3| \leq \mathsf{negl}(\lambda), \qquad p_3 = \frac{1}{2}.$$

Conclusion.

$$|p_0 - \tfrac{1}{2}| \leq |p_0 - p_1| + |p_1 - p_2| + |p_2 - p_3| \leq \mathsf{negl}(\lambda).$$

3 Formal Security Analysis

3.1 BAN Logic Analysis

BAN logic is employed to analyze the authentication and key-agreement beliefs established by GroupTEE. The purpose of this analysis is not to verify algebraic correctness of the key computation, but to reason about belief establishment and mutual authentication.

The analysis proceeds in four stages: (1) formalization of security goals, (2) idealization of protocol messages, (3) definition of initial assumptions, (4) derivation using BAN inference rules.

We emphasize that BAN logic abstracts Schnorr signatures as ideal authentication primitives. Since all signed messages in GroupTEE include the session identifier *sid*, freshness is bound to the session context.

The following BAN inference rules are used.

Jurisdiction Rule:

$$\frac{P \mid\equiv Q \Rightarrow X \quad P \mid\equiv Q \mid\equiv X}{P \mid\equiv X} \tag{1}$$

Message-Meaning Rule (Signature Form):

$$\frac{P \mid\equiv \mathrm{pub}_Q \leftrightarrow Q \quad P \lhd \{X\}_{sk_Q}}{P \mid\equiv Q \mid\sim X} \tag{2}$$

Nonce-Verification Rule:

$$\frac{P \mid\equiv \#(X) \quad P \mid\equiv Q \mid\sim X}{P \mid\equiv Q \mid\equiv X} \tag{3}$$

Formal Proof Goal The goal of GroupTEE is for all legitimate participants $P_1, \ldots, P_n$ to achieve mutual authentication and to agree on the same fresh group session key after two rounds of interaction. Since GroupTEE derives a single group key GK shared by all members in the authenticated participant list, we specify the following BAN-logic goals for any two distinct members P_i and P_j $(i \neq j)$:

Goal1 (Key belief). $P_i \mid\equiv \big(P_i \overset{GK}{\longleftrightarrow} P_j\big)$.

Goal2 (Key belief). $P_j \mid\equiv \big(P_i \overset{GK}{\longleftrightarrow} P_j\big)$.

Goal3 (Mutual belief). $P_i \mid\equiv P_j \mid\equiv \big(P_i \overset{GK}{\longleftrightarrow} P_j\big)$.

Goal4 (Mutual belief). $P_j \mid\equiv P_i \mid\equiv \big(P_i \overset{GK}{\longleftrightarrow} P_j\big)$.

In addition, we require freshness of the established group key to be bound to the session context:

Goal5 (Key freshness). $P_i \mid\equiv \#(GK)$ and $P_j \mid\equiv \#(GK)$.

Idealized Protocol Let $\{X\}_{sk_i}$ denote a Schnorr signature generated using P_i's private key. The session identifier sid is included in all signed messages and serves as the freshness indicator of the protocol session.

The idealized messages are:

Message 1 (First Round).

$$P_i \rightarrow P_{i\pm1} : P_{i\pm1} \lhd \{ID_i, R_i, sid\}_{sk_i}$$

Message 2 (Second Round Broadcast).

$$P_i \rightarrow P_j \ (\forall j \neq i) : P_j \lhd \{ID_i, X_i, R_i, sid\}_{sk_i}$$

Initial Assumptions Before protocol execution, we formalize the trust relationships, freshness conditions, and key-derivation properties that are assumed to hold among honest participants.

A1: $P_i \mid\equiv CA \Rightarrow (\mathrm{pub}_k \leftrightarrow ID_k)\ (\forall k)$

A2: $P_i \mid\equiv CA \mid\equiv (\text{pub}_k \leftrightarrow ID_k)\ (\forall k)$
A3: $P_i \mid\equiv \#(sid)$
A3': $P_i \mid\equiv \#(sid) \Rightarrow \#(ID_k, R_k, sid)\ \wedge\ \#(ID_k, X_k, R_k, sid)\ (\forall k)$
A4: $P_i \mid\equiv \#(R_k)$
A5: $P_i \mid\equiv \#(X_k)$
A6: $P_i \mid\equiv GK = \mathsf{KDF}(f(R_1, \ldots, R_n, X_1, \ldots, X_n, sid))$

Derivation Consider two arbitrary participants P_i and P_j $(i \neq j)$.

Step 1: Public-key binding.

From A1 and A2 and applying the Jurisdiction rule, participant P_j believes:

$$P_j \mid\equiv (\text{pub}_i \leftrightarrow ID_i).$$

Step 2: First-round authentication.

Upon receiving Message 1 and applying the Message-Meaning rule:

$$P_j \mid\equiv P_i \mid\sim (ID_i, R_i, sid).$$

Since sid is fresh (A3), by Nonce Verification:

$$P_j \mid\equiv P_i \mid\equiv (ID_i, R_i).$$

Thus P_j believes R_i originates from P_i in this session.

Step 3: Second-round authentication.

Upon receiving Message 2 and applying Message-Meaning:

$$P_j \mid\equiv P_i \mid\sim (ID_i, X_i, R_i, sid).$$

Using freshness of sid again:

$$P_j \mid\equiv P_i \mid\equiv (ID_i, X_i, R_i).$$

Therefore, P_j believes that X_i and R_i are fresh and generated by P_i.

Step 4: Key agreement belief.

Since all participants derive the session key from the same authenticated set $\{(ID_k, R_k, X_k, sid)\}_{k=1}^{n}$ and believe in the correctness of the key-derivation function (A6), it follows that:

$$P_i \mid\equiv P_i \overset{GK}{\longleftrightarrow} P_j,$$

$$P_j \mid\equiv P_i \overset{GK}{\longleftrightarrow} P_j.$$

Furthermore, since each party believes that the other party believes in the same authenticated parameters, we obtain:

$$P_i \mid\equiv P_j \mid\equiv P_i \overset{GK}{\longleftrightarrow} P_j,$$

$$P_j \mid\equiv P_i \mid\equiv P_i \overset{GK}{\longleftrightarrow} P_j.$$

Therefore, under the stated assumptions, GroupTEE achieves mutual authentication and shared session-key belief among all honest participants. Each party believes that the established key is fresh, authenticated, and shared exclusively with legitimate group members.

3.2 Formal Analysis Tool

Scyther is employed to perform symbolic verification of GroupTEE under the Dolev–Yao adversary model. In the Scyther specification, Schnorr signatures are abstracted as ideal digital signatures $\{m\}_{\mathrm{sk}}$ that are publicly verifiable via the corresponding public key, captured by the relation $\mathrm{inversekeys}(\mathrm{pk}, \mathrm{sk})$. The session identifier *sid*, which abstracts the session context in the protocol, is included in the signed payload to enforce session binding and to prevent replay across sessions.

We specify standard authentication claims Alive, Weakagree, Nisagree, Nisyncn for each role and a secrecy claim on an abstract session key derived from the authenticated broadcast parameters of both rounds. Fig. 1 illustrates the Scyther-based formal modeling of the GroupTEE protocol, and the results show that all specified security claims are successfully satisfied within the given bounded analysis scope, as illustrated in Fig. 2.The authentication- and synchronization-related claims, including Alive, Weakagree, Niagree, and Nisynch, are all marked as Ok, indicating that the protocol is resilient to replay, impersonation, and desynchronization attacks within the current model and search bounds.

References

1. Bellare, M., Garay, J.A., Rabin, T.: Fast batch verification for modular exponentiation and digital signatures. In: International conference on the theory and applications of cryptographic techniques. pp. 236–250. Springer (1998)
2. Pointcheval, D., Stern, J.: Security proofs for signature schemes. In: International conference on the theory and applications of cryptographic techniques. pp. 387–398. Springer (1996)

```
hashfunction H;
const pkA, pkB, pkC: Function;
secret skA, skB, skC: Function;
inversekeys (pkA,skA);
inversekeys (pkB,skB);
inversekeys (pkC,skC);

protocol SEAGKAP(A, B, C)
{
  role A
  {
    fresh rA: Nonce;
    fresh RA: Nonce;
    var RB: Nonce;
    var RC: Nonce;
    fresh sid: Nonce;
    send_1(A, B, A, RA, sid, {A,RA,sid}skA);
    send_2(A, C, A, RA, sid, {A,RA,sid}skA);
    recv_3(B, A, B, RB, sid, {B,RB,sid}skB);
    recv_5(C, A, C, RC, sid, {C,RC,sid}skC);

    fresh XA: Nonce;
    var XB: Nonce;
    var XC: Nonce;
    send_7(A, B, A, XA, sid, {A,XA,sid}skA);
    send_8(A, C, A, XA, sid, {A,XA,sid}skA);
    recv_9(B, A, B, XB, sid, {B,XB,sid}skB);
    recv_11(C, A, C, XC, sid, {C,XC,sid}skC);

    claim(A, Secret, H(rA, RC, XA, XB));
    claim(A, Alive);
    claim(A, Weakagree);
    claim(A, Niagree);
    claim(A, Nisynch);
  }

  role B
  {
    fresh RB: Nonce;
    fresh rB: Nonce;
    var RA: Nonce;
    var RC: Nonce;
    var sid: Nonce;
    recv_1(A, B, A, RA, sid, {A,RA,sid}skA);
    send_3(B, A, B, RB, sid, {B,RB,sid}skB);
    send_4(B, C, B, RB, sid, {B,RB,sid}skB);
    recv_6(C, B, C, RC, sid, {C,RC,sid}skC);

    fresh XB: Nonce;
    var XA: Nonce;
    var XC: Nonce;
    recv_7(A, B, A, XA, sid, {A,XA,sid}skA);
    send_9(B, A, B, XB, sid, {B,XB,sid}skB);
    send_10(B, C, B, XB, sid, {B,XB,sid}skB);
    recv_12(C, B, C, XC, sid, {C,XC,sid}skC);

    claim(B, Secret, H(rB, RA, XB, XC));
    claim(B, Alive);
    claim(B, Weakagree);
    claim(B, Niagree);
    claim(B, Nisynch);
  }
  role C
  {
    fresh rC: Nonce;
    fresh RC: Nonce;
    var RA: Nonce;
    var RB: Nonce;
    var sid: Nonce;
    recv_2(A, C, A, RA, sid, {A,RA,sid}skA);
    recv_4(B, C, B, RB, sid, {B,RB,sid}skB);
    send_5(C, A, C, RC, sid, {C,RC,sid}skC);
    send_6(C, B, C, RC, sid, {C,RC,sid}skC);

    fresh XC: Nonce;
    var XA: Nonce;
    var XB: Nonce;
    recv_8(A, C, A, XA, sid, {A,XA,sid}skA);
    recv_10(B, C, B, XB, sid, {B,XB,sid}skB);
    send_11(C, A, C, XC, sid, {C,XC,sid}skC);
    send_12(C, B, C, XC, sid, {C,XC,sid}skC);
    claim(C, Secret, H(rC, RB, XC, XA));
    claim(C, Alive);
    claim(C, Weakagree);
    claim(C, Niagree);
    claim(C, Nisynch);
  }
}
```

Fig. 1. Scyther-based formal model of GroupTEE

Scyther results : verify

Claim				Status	Comments
SEAGKAP	A	SEAGKAP,A1	Secret H(rA,RC,XA,XB)	Ok	No attacks within bounds.
		SEAGKAP,A2	Alive	Ok	No attacks within bounds.
		SEAGKAP,A3	Weakagree	Ok	No attacks within bounds.
		SEAGKAP,A4	Niagree	Ok	No attacks within bounds.
		SEAGKAP,A5	Nisynch	Ok	No attacks within bounds.
	B	SEAGKAP,B1	Secret H(rB,RA,XB,XC)	Ok	No attacks within bounds.
		SEAGKAP,B2	Alive	Ok	No attacks within bounds.
		SEAGKAP,B3	Weakagree	Ok	No attacks within bounds.
		SEAGKAP,B4	Niagree	Ok	No attacks within bounds.
		SEAGKAP,B5	Nisynch	Ok	No attacks within bounds.
	C	SEAGKAP,C1	Secret H(rC,RB,XC,XA)	Ok	No attacks within bounds.
		SEAGKAP,C2	Alive	Ok	No attacks within bounds.
		SEAGKAP,C3	Weakagree	Ok	No attacks within bounds.
		SEAGKAP,C4	Niagree	Ok	No attacks within bounds.
		SEAGKAP,C5	Nisynch	Ok	No attacks within bounds.

Done.

Fig. 2. Scyther-based formal analysis result of GroupTEE